Spectral density estimation

Given a discrete signal $x[n]$, $n = 0, \dots, N-1$, its Discrete Fourier Transform (DFT) is

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}, \quad k = 0, \dots, N-1.$$

Periodogram

An estimate of the signal's power distribution over frequency

$$P_x(f_k) = \frac{1}{N} \left| X[k] \right|^2, \quad f_k = \frac{k}{N} f_s.$$

Power spectral density:

$$P = \lim_{T \to \infty} \frac{1}{T} \int_{t_0 - T/2}^{t_0 + T/2} \left| x(t) \right|^2 dt$$

$$x_T(t) = x(t) w_T(t)$$

$$P = \lim_{T \to \infty} \frac{1}{T} \int_{-\infty}^{\infty} \left| x_T(t) \right|^2 dt$$

$$\text{Parseval's theorem} \quad - \quad P = \lim_{T \to \infty} \frac{1}{T} \int_{-\infty}^{\infty} \left| \hat{x}_T(f) \right|^2 df$$

$$S_{xx}(f) = \lim_{N \to \infty} \frac{(\Delta t)^2}{T} \left| \sum_{n=-N}^{N} x_n e^{-i2\pi fn\,\Delta t} \right|^2$$

The Bartlett method

Issues:

1. long signals ($N \gg 1$) $\to$ high resolution DFT ($\Delta f = f_s/N \ll 1$)
2. high computational cost ($N \log_2 N$)
3. high variance DFT (noisy spectrum)

Bartlett method

Subdivide the observed data record into M nonoverlapping segments, find the periodogram (spectrum) of each segment, and finally evaluate the average of the periodograms obtained.

$$x_m[n] = x[n + (m-1)L] \ , \quad 0 \leqslant n \leqslant L-1 \quad \to \quad I_m[k] = \frac{1}{L}\left|\sum_{n=0}^{L-1} x_m[n] e^{2\pi ikn/L}\right|^2$$

$$S_B[k] = \frac{1}{M}\sum_{m=1}^{M} I_m[k] \qquad \text{Bartlett estimator}$$

Advantage:

1. low resolution DFT ($\Delta f = f_s/L$)
2. lower computational cost ($\sim M \cdot L \log_2 L = N \log_2 L$)
3. lower variance (smoother spectrum)

New issues:

1. shorter segments $\to$ high leakage (strong bias) (see effects of windowing)
2. if short signal $\to$ few segments $\to$ poor statistics (high variance)

Variance-bias trade-off

- Longer segments: high variance, low bias
- Short segments: low variance, high bias

Welch's method

Similar to the Bartlett but segments can be overlapping and each segment is **tapered(windowed)** to reduce spectral leakage.

Taper = window with smoothly decaying edges (for reducing leakage due to truncation)

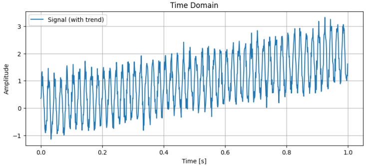
$$x_m[n] = x[n + (m-1)D] \ , \quad 0 \leqslant n \leqslant L-1 \ , \ D \leqslant L \quad \rightarrow \quad \tilde{I}_m[k] = \frac{1}{L}\left|\sum_{n=0}^{L-1} w[n] x_m[n] e^{2\pi i k n/L}\right|^2$$

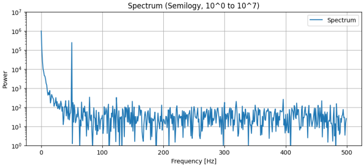
$w[n]$ is a window of duration L

$$S_W[k] = \frac{1}{M}\sum_{m=1}^{M} \tilde{I}_m[k] \qquad \text{Welch estimator}$$

The window is normalized: $\sum_{n=0}^{L-1} w^2[n] = L$

Issue: trends with time scale beyond the window size pollute the spectrum at low frequencies





Explanation

Shifting a signal away from zero adds a Dirac delta at $f = 0$ because the Fourier transform of a constant is

$$\mathcal{F}[a] = a\int_{-\infty}^{\infty} e^{-2\pi i f t} \mathrm{d}t = a\delta(f) \quad \text{so} \quad \mathcal{F}(x+a) = \mathcal{F}(x) + a\delta(f)$$

Detrending

Before windowing **detrending** is applied.

Detrending is the process of removing a low-order trend (typically the mean or a linear component) from a signal before spectral analysis

$$x_d[n] = x[n] - \hat{x}_{\text{trend}}[n].$$

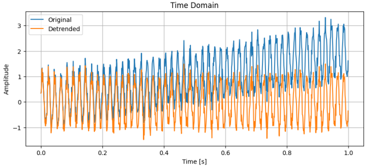
Common choices:

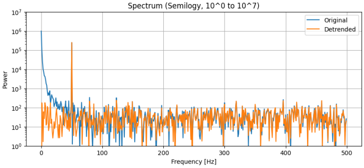
$$\text{Mean removal: } x_d[n] = x[n] - \frac{1}{N}\sum_{n=0}^{N-1} x[n],$$

$$\text{Linear detrending: } x_d[n] = x[n] - (an + b),$$

where $an + b$ is the best-fit line.

Advantage: reduces artificial low-frequency components that would otherwise dominate the spectrum.





Welch's method: summary

Given a signal $x[n]$:

1. **Segmentation:** Divide the signal into overlapping segments of length L:
$$x_k[n], \qquad n = 0, \ldots, L-1, \quad k = 1, \ldots, M.$$
2. **Detrending:** Remove the mean or a low-order trend from each segment:
$$x_{k,d}[n] = x_k[n] - \hat{x}_{\mathrm{trend},k}[n].$$
3. **Windowing:** Multiply each segment by a window function:
$$x_{k,w}[n] = x_{k,d}[n]\, w[n].$$
4. **Zero padding:** Extend the windowed segment to an FFT length $N_{\mathrm{FFT}} \geq L$:
$$x_{k,p}[n] = \begin{cases} x_{k,w}[n], & 0 \leq n < L, \\ 0, & L \leq n < N_{\mathrm{FFT}}. \end{cases}$$
This refines the frequency grid but does not improve true frequency resolution.
5. **DFT:** Compute the DFT of each padded, windowed segment:
$$X_k[m] = \sum_{n=0}^{N_{\mathrm{FFT}}-1} x_{k,p}[n] e^{-j2\pi mn/N_{\mathrm{FFT}}}.$$
6. **PSD normalization:**
$$P_k(f_m) = \frac{1}{f_s \sum_{n=0}^{L-1} w^2[n]} \left|X_k[m]\right|^2, \qquad f_m = \frac{m}{N_{\mathrm{FFT}}} f_s.$$
7. **Averaging:**
$$\hat{S}_{xx}(f_m) = \frac{1}{M} \sum_{k=1}^{M} P_k(f_m).$$
8. **One-sided correction for real signals:** Keep only $0 \leq f \leq f_s/2$ and double the interior frequency bins:
$$\hat{S}_{xx}^{(1)}(f_m) = \begin{cases} \hat{S}_{xx}(f_m), & f_m = 0 \text{ or } f_m = f_s/2, \\ 2\hat{S}_{xx}(f_m), & 0 < f_m < f_s/2. \end{cases}$$

Welch's method estimates the PSD by averaging normalized, windowed segment periodograms.

Averaging reduces variance, windowing reduces leakage, detrending reduces artificial low-frequency dominance, and zero padding refines the plotted frequency grid.

Remark

The *one-sided* spectrum is calculated only for non-negative frequencies because it assumes the signal to be a real function and for a real $x(t)$

$$X(-\omega) = X^*(\omega) \quad \rightarrow \quad |X(-\omega)|^2 = |X^*(\omega)|^2 = |X(\omega)|^2 \ .$$

From the set of frequencies $f_k = kf_s/N$ where $k = \{-N/2, -N/2+1, \ldots, -1, 0, 1, \ldots N/2-1\}$ the amplitudes for $k = \pm 1$ and $k = \pm 2$ and $\ldots$ $k = \pm(N/2-1)$ will appear twice while for $k = 0$ and $k = N/2$ (actually $k = -N/2$) will appear once. Therefore we keep the different ones: $k = \{0, 1, \ldots, N/2\}$ i.e. $(N/2+1)$ elements.

Laplace transform

Most common *integral transform* (beside the Fourier transform). Applications in mathematics, physics and engineering.

Integral transformation

$$F(s) = \mathcal{T}[f] = \int_{-\infty}^{+\infty} K(s,t) f(t) dt \ ,$$

where $K(s,t)$ is the *kernel* of the transformatoin.

$$F(s) = \mathcal{T}[f], \quad \text{és} \quad G(s) = \mathcal{T}[g] \ ,$$

Linearity

$$\mathcal{T}[c_1 f(t) + c_2 g(t)] = c_1 F(s) + c_2 G(s) \ .$$

In some cases there is also an *inverse* transformation:

$$f(t) = \mathcal{T}^{-1}[F] = \int_{-\infty}^{+\infty} K^{-1}(s,t) F(s) ds \ ,$$

where $K^{-1}(s,t)$ is the transformation's *inverse kernel*.

The Laplace transform is defined for functions with positive real input and its kernel is

$$K(s,t) \equiv e^{-st} \ .$$

Laplace transform

$$F(s) = \mathcal{L}[f] = \int_0^\infty e^{-st} f(t) dt \ , \quad s \in \mathbb{C} \ , \quad F : \mathbb{C} \to \mathbb{C}$$

$s = \sigma + i\omega$ - **complex frequency**

1

$$\mathcal{L}[1] = \frac{1}{s}, \qquad s > 0;$$

2

$$\mathcal{L}[e^{at}] = \frac{1}{s-a}, \qquad s > 0;$$

Proof:

$$\mathcal{L}[e^{at}] = \int_0^\infty e^{-st} e^{at}\, dt = \int_0^\infty e^{(a-s)t}\, dt = \left. \frac{e^{(a-s)t}}{a-s} \right|_0^\infty = \frac{1}{s-a}.$$

3

$$\mathcal{L}[\cos at] = \frac{s}{s^2+a^2}, \quad \mathcal{L}[\sin at] = \frac{a}{s^2+a^2}.$$

Proof:
Based on the previous property

$$\mathcal{L}[e^{iat}] = \mathcal{L}[\cos at] + i\mathcal{L}[\sin at] = \frac{1}{s-ia} = \frac{s+ia}{s^2+a^2}. \square$$

4.

$$\mathcal{L}[t^p] = \frac{\Gamma(p+1)}{s^{p+1}}, \qquad s > 0;$$

Proof:

$$\mathcal{L}[t^p] = \int_0^\infty e^{-st} t^p dt = \frac{1}{s^{p+1}} \int_0^\infty e^{-u} u^p du = \frac{1}{s^{p+1}} \Gamma(p+1).$$

$$\mathcal{L}[t^n] = \frac{n!}{s^{n+1}}$$

5.

$$\mathcal{L}[e^{ct} f(t)] = F(s-c)$$

Proof:

$$\mathcal{L}[e^{ct} f(t)] = \int_0^\infty e^{-st} e^{ct} f(t) dt = \int_0^\infty e^{-(s-c)t} f(t) dt = F(s-c).$$

6

$$\mathcal{L}[e^{ct}\sin at] = \frac{a}{(s-c)^2+a^2}$$

$$\mathcal{L}[e^{ct}\cos at] = \frac{s-c}{(s-c)^2+a^2}$$

7

$$\mathcal{L}[\sinh at] = \frac{a}{s^2-a^2}\ , \qquad \mathcal{L}[\cosh at] = \frac{s}{s^2-a^2}$$

Proof:

$$\mathcal{L}[\sinh at] = \frac{1}{2}\left(\mathcal{L}[e^{at}] - \mathcal{L}[e^{-at}]\right) = \frac{1}{2}\left(\frac{1}{s-a} - \frac{1}{s+a}\right) = \frac{a}{s^2-a^2}\,.$$

$$\mathcal{L}[\cosh at] = \frac{1}{2}\left(\mathcal{L}[e^{at}] + \mathcal{L}[e^{-at}]\right) = \frac{1}{2}\left(\frac{1}{s-a} + \frac{1}{s+a}\right) = \frac{s}{s^2-a^2}\,.$$

8

$$\mathcal{L}[f(ct)] = \frac{1}{c}F\left(\frac{s}{c}\right)$$

Proof:

$$\mathcal{L}[f(ct)] = \int_0^\infty e^{-st}f(ct)dt = \frac{1}{c}\int_0^\infty e^{-\frac{s}{c}u}f(u)du = \frac{1}{c}F\left(\frac{s}{c}\right)\,.$$

9

$$\mathcal{L}[\delta(t)] = 1\ , \quad \mathcal{L}[\delta(t-t_0)] = e^{-st_0}\ .$$

10

$$\mathcal{L}[H(t-t_0)] = \frac{e^{-st_0}}{s} \ ,$$

where $H(t)$ is the *Heaviside step function*:

$$H(t) = \begin{cases} 0 \ , & t < 0 \ , \\ 1 \ , & t \geq 0 \ . \end{cases}$$

11

$$\mathcal{L}[H(t-a) - H(t-b)] = \frac{e^{-as} - e^{-bs}}{s}$$

12

$$\mathcal{L}[(-t)^n f(t)] = F^{(n)}(s) \ .$$

Proof:

$$\frac{\mathrm{d}F(s)}{\mathrm{d}s} = \int_0^\infty (-t)e^{-st} f(t)dt = \mathcal{L}[-tf(t)] \ .$$

Examples

$$\mathcal{L}[t \sinh at] = -\frac{d}{ds}\left(\frac{a}{s^2 - a^2}\right) = \frac{2as}{(s^2 - a^2)^2}$$

$$\frac{1}{(s-a)^2} = -\frac{\mathrm{d}}{\mathrm{d}s}\left(\frac{1}{s-a}\right) = \mathcal{L}[te^{at}]$$

$$\frac{1}{(s-a)^{k+1}} = \frac{(-1)^k}{k!}\frac{\mathrm{d}^k}{\mathrm{d}s^k}\left(\frac{1}{s-a}\right) = \mathcal{L}[\frac{1}{k!}t^k e^{at}]$$

Convolution sum and Laplace transform

The convolution of f and g

$$f * g \equiv \int_0^t f(\tau)g(t-\tau)d\tau \ , \quad f, g : \mathbb{R}^+ \to \mathbb{R} \ .$$

Remark:

You can look at the above definition as one with integration over the whole real axis and the limits change due to the fact that the participating functions should be zero over the negative half axis:

$$f * g \equiv \int_{-\infty}^{\infty} f(\tau)g(t-\tau)d\tau \ , \quad f, g : \mathbb{R} \to \mathbb{R} \ .$$

$\tau < 0 \to f = 0 \ , \tau > t \to g = 0.$

Properties

$$\begin{aligned} f * g &= g * f, \qquad \textit{commutativity}; \\ f * (g_1 + g_2) &= f * g_1 + f * g_2, \qquad \textit{distributivity}; \\ (f * g) * h &= f * (g * h), \qquad \textit{associativity}; \end{aligned}$$

$$\mathcal{L}[f * g] = \mathcal{L}[f] \cdot \mathcal{L}[g].$$

Proof

$$f * g = \int_0^t f(\tau)g(t - \tau)d\tau = \int_0^\infty f(\tau)g(t - \tau)d\tau$$

as $g(t - \tau) = 0$ ha $\tau > t$. Applying the change of variables $t - \tau = u, \quad t \longrightarrow u, \quad dt = du$:

$$\mathcal{L}[f * g] = \int_0^\infty e^{-st} \int_0^\infty f(\tau)g(t - \tau)d\tau dt = \int_0^\infty \int_0^\infty e^{-s(\tau+u)} f(\tau)g(u)d\tau du = .$$

$$= \mathcal{L}[f]\mathcal{L}[g]$$

Examples: find the inverse Laplace transforms

1. $$Y(s) = \frac{3s - 1}{s^2 + 4}.$$

   Splitting it into terms

   $$Y(s) = 3\frac{s}{s^2 + 4} - \frac{1}{2}\frac{2}{s^2 + 4}$$

   whence

   $$y = 3\cos 2t - \frac{1}{2}\sin 2t\ .$$

2. $$Y(s) = \frac{s - 1}{s^2 - s - 2}\ .$$

   $$Y(s) = \frac{s - 1}{(s + 1)(s - 2)} = \frac{A}{s + 1} + \frac{B}{s - 2} = \frac{2/3}{s + 1} + \frac{1/3}{s - 2}\ .$$

   Using the property $F(s - c) \longleftrightarrow e^{ct}f(t)$:

   $$y = \frac{2}{3}e^{-t} + \frac{1}{3}e^{2t}$$

3. $$H(s) = \frac{a}{s^2(s^2 + a^2)}$$

   We look at $H(s)$ as a product between s^{-2} és $a/(s^2 + a^2)$ which are the Laplace transforms of t and $\sin at$ respectively

   $$h(t) = \int_0^t (t - \tau)\sin a\tau\, d\tau = \frac{at - \sin at}{a^2}$$

   or because of the commutativity

   $$h(t) = \int_0^t \tau \sin a(t - \tau) d\tau.$$